

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
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COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.
3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based

on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

1) ANSWER:

Network Throughput and Packet Loss Calculation

Problem Understanding

In a Cisco Packet Tracer network simulation, the total data received is 8000 bits in 4 seconds. A total of 200 packets are sent, out of which 180 packets are received. The objective is to calculate the network throughput and packet loss percentage, classify the network as either Efficient or Inefficient, and provide a complete performance summary.

Calculation

The throughput is calculated using:

Throughput = Total Data Received / Time

Therefore,

Throughput = $8000 / 4 = 2000$ bps

The number of lost packets is:

Lost Packets = Packets Sent – Packets Received

Lost Packets = $200 - 180 = 20$ packets

Packet loss percentage is:

Packet Loss = $(\text{Lost Packets} / \text{Packets Sent}) \times 100$

Packet Loss = $(20 / 200) \times 100 = 10\%$

The network is classified as Efficient only when throughput is greater than 1500 bps AND packet loss is less than 10%. Here, throughput is 2000 bps, which satisfies the first condition. However, packet loss is exactly 10%, not less than 10%. Therefore, the network is classified as Inefficient.

Pseudocode

ALGORITHM Network_Performance

BEGIN

 INPUT totalDataReceived

 INPUT time

 INPUT packetsSent

 INPUT packetsReceived

 IF time <= 0 THEN

 DISPLAY "Error: Time must be greater than zero"

 STOP

 END IF

 IF packetsSent <= 0 THEN

 DISPLAY "Error: Packets sent must be greater than zero"

 STOP

 END IF

IF packetsReceived < 0 OR packetsReceived > packetsSent THEN

 DISPLAY "Error: Invalid packets received"

 STOP

END IF

throughput $\leftarrow$ totalDataReceived / time

lostPackets $\leftarrow$ packetsSent - packetsReceived

packetLoss $\leftarrow$ (lostPackets / packetsSent) $\times$ 100

DISPLAY "Throughput = ", throughput, " bps"

DISPLAY "Packet Loss = ", packetLoss, "%"

DISPLAY "Packets Lost = ", lostPackets

IF throughput > 1500 AND packetLoss < 10 THEN

 DISPLAY "Network Classification = Efficient"

ELSE

 DISPLAY "Network Classification = Inefficient"

END IF

DISPLAY "Network Performance Summary"

DISPLAY "Throughput: ", throughput, " bps"

DISPLAY "Packet Loss: ", packetLoss, "%"

DISPLAY "Classification: Efficient/Inefficient"

END

Expected Result

Throughput = 2000 bps

Packet Loss = 10%

Packets Lost = 20

Network Classification = Inefficient

Explanation of Algorithm

The algorithm first accepts the network measurements as input and validates the time and packet values. It then calculates throughput using the amount of data received and the time taken. Packet loss is calculated by finding the difference between packets sent and received. Finally, an IF-ELSE condition compares the calculated values with the given efficiency criteria and displays the final network performance.

This satisfies the rubric because it identifies all inputs and constraints, provides a complete algorithm, uses conditions appropriately, and handles invalid input cases. The rubric specifically expects a complete and logically sound solution at Level 4

2) ANSWER:

Bandwidth Utilization Monitoring and Congestion Prevention

Problem Understanding

An organization has multiple network links connecting its branches. The administrator needs to continuously monitor the bandwidth utilization of each link and identify overloaded links. The bandwidth values should be collected periodically and stored in an array or list

Proposed Approach

The algorithm uses an array to store the utilization values of all network links. A `FOR` loop is used to check every link during each monitoring cycle. A `WHILE` loop ensures that monitoring continues periodically.

When the utilization is greater than 80%, the link is considered overloaded. The administrator is then advised to move traffic to a backup link.

Pseudocode

ALGORITHM Bandwidth_Monitoring

BEGIN

 INPUT numberOfLinks

 IF numberOfLinks <= 0 THEN

 DISPLAY "Error: Invalid number of links"

 STOP

 END IF

 CREATE utilization[numberOfLinks]

 WHILE TRUE DO

 DISPLAY "Starting bandwidth monitoring"

 FOR i ← 1 TO numberOfLinks DO

 INPUT utilization[i]

 IF utilization[i] < 0 OR utilization[i] > 100 THEN

 DISPLAY "Invalid utilization value for Link ", i

 CONTINUE

 END IF

 DISPLAY "Link ", i, " Utilization = ",

 utilization[i], "%"

 IF utilization[i] > 80 THEN

 DISPLAY "Warning: Link ", i, " is overloaded"

```
        DISPLAY "Recommend switching traffic to backup link"

    ELSE

        DISPLAY "Link ", i, " is operating normally"

    END IF

END FOR

WAIT monitoringInterval

END WHILE

END
```

Explanation

The algorithm continuously monitors every network link. The `FOR` loop ensures that each link is checked, while the `WHILE` loop allows monitoring to continue at regular intervals.

The 80% utilization threshold is used to identify potentially congested links. When utilization exceeds this value, the algorithm generates a warning and recommends moving traffic to a backup link. This prevents a highly utilized link from becoming a major congestion point.

The use of periodic monitoring also allows the administrator to respond to changing traffic conditions instead of relying on a single measurement.

How It Improves Bandwidth Management

This algorithm improves bandwidth management by identifying overloaded links before network performance becomes severely affected. Traffic can be redirected to available backup links, which helps reduce congestion and maintain better network performance.

It also satisfies the rubric requirement for effective use of loops and conditions, since the algorithm uses a `WHILE` loop for continuous monitoring, a `FOR` loop for checking each link, and `IF-ELSE` conditions for detecting overloaded links.

3)

ANSWER:

Best Communication Path Based on Link Quality

Problem Understanding

A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay and utilization, and these values change throughout the day.

The objective is to continuously identify the best communication path between headquarters and each branch. Instead of selecting a path only according to physical distance, the algorithm should consider the overall quality of the link.

The algorithm must periodically collect bandwidth, delay and utilization values, calculate a link quality score, select the path with the highest score, identify overloaded or failed links, automatically select an alternate path, and generate an alert when necessary.

Link Quality Concept

A good communication path should have:

- High bandwidth
- Low delay
- Low utilization

Therefore, a quality score can combine these three metrics.

A suitable weighted score is:

Link Quality Score = $(0.5 \times \text{Bandwidth Score}) + (0.3 \times \text{Delay Score}) + (0.2 \times \text{Utilization Score})$

Here, bandwidth is given the highest importance because higher bandwidth generally supports better data transmission. Lower delay and lower utilization also improve the quality of the path.

Pseudocode

ALGORITHM Best_Communication_Path

BEGIN

 INPUT numberOfLinks

 IF numberOfLinks <= 0 THEN

 DISPLAY "Error: Invalid number of links"

 STOP

 END IF

 CREATE bandwidth[numberOfLinks]

 CREATE delay[numberOfLinks]

 CREATE utilization[numberOfLinks]

 CREATE status[numberOfLinks]

 CREATE score[numberOfLinks]

```

WHILE TRUE DO

    bestScore ← -1

    bestPath ← NONE

    FOR i ← 1 TO numberOfLinks DO

        INPUT bandwidth[i]

        INPUT delay[i]

        INPUT utilization[i]

        INPUT status[i]

        IF status[i] = "FAILED" THEN

            DISPLAY "Link ", i, " has failed"

            score[i] ← -1

            CONTINUE

        END IF

        IF utilization[i] > 80 THEN

            DISPLAY "Warning: Link ", i, " is overloaded"

            score[i] ← -1

            CONTINUE

        END IF

        bandwidthScore ← NormalizeBandwidth(bandwidth[i])

        delayScore ← NormalizeDelay(delay[i])

        utilizationScore ← 100 - utilization[i]

        score[i] ←

            (0.5 × bandwidthScore) +

            (0.3 × delayScore) +

            (0.2 × utilizationScore)

        IF score[i] > bestScore THEN

            bestScore ← score[i]

            bestPath ← i

        END IF

    END FOR

    IF bestPath = NONE THEN

        DISPLAY "No suitable path available"

```



```
    DISPLAY "ALERT: Network connectivity problem"
```

```
ELSE
```

```
    DISPLAY "Best Communication Path = Link ", bestPath
```

```
    DISPLAY "Link Quality Score = ", bestScore
```

```
END IF
```

```
WAIT monitoringInterval
```

```
END WHILE
```

```
END
```

Explanation

The algorithm continuously collects the current network conditions. First, it checks whether a link has failed. Failed links are immediately excluded from consideration. It then checks utilization. If utilization exceeds 80%, the link is treated as overloaded and is also excluded from the current path selection.

For every valid link, the algorithm calculates a quality score using bandwidth, delay and utilization. The path with the highest score is selected as the best communication path.

Because the monitoring process is repeated periodically, the selected path can change when network conditions change. If the current best path becomes overloaded or fails, another suitable path can automatically be selected.

Benefits

Network Reliability: Failed links are avoided and alternate paths are available.

Routing Efficiency: The algorithm considers multiple performance metrics rather than only distance.

Fault Tolerance: When a link fails or becomes overloaded, another path can be selected.

Adaptability: The algorithm continuously updates its decision based on changing network conditions.

Administrator Awareness: Alerts are generated when no suitable path is available.

This directly addresses the question's requirements for reliability, routing efficiency, fault tolerance and adaptation to changing conditions.